
1. Project Introduction

1.1 Overview of the Project

The **ReqRes API Testing Project using Postman** is a software testing project focused on validating RESTful APIs provided by the ReqRes service (<https://reqres.in>). This project demonstrates how API testing is performed in real-world QA environments by verifying that backend services work correctly before they are used in frontend applications.

APIs act as a communication bridge between client applications and servers. Therefore, testing APIs is a critical part of software quality assurance. In this project, different HTTP methods such as **GET, POST, PUT, PATCH, and DELETE** are used to test various functionalities like user authentication, user data retrieval, user creation, updating user details, and deleting users.

1.2 Purpose of the Project

The main purpose of this project is to understand and implement **manual API testing using Postman** and to ensure that all API endpoints are working as expected.

This project helps to:

- Validate API request and response behavior
- Ensure correct status codes are returned
- Verify response data accuracy
- Identify bugs or unexpected behavior in APIs
- Build real-time API testing skills used in QA industry

It also simulates a real-time testing environment where testers ensure backend services are stable and reliable.

1.3 What is ReqRes API?

ReqRes (<https://reqres.in>) is a free online REST API used for testing and learning purposes. It provides dummy data for users, login functionality, and CRUD operations.

This API is widely used by beginners and professionals because:

- It is free and publicly accessible
- It supports all major HTTP methods
- It provides predictable responses

- It is ideal for learning API automation and manual testing

In this project, ReqRes is used as the test environment instead of a real production system.

1.4 Tools Used in This Project

Several tools and technologies are used to perform and manage API testing:

Postman

Postman is a widely used API testing tool that allows testers to send requests to APIs and analyze responses. It provides features like collections, environments, test scripts, and automated execution.

Newman

Newman is the command-line tool for Postman. It allows execution of Postman collections from the terminal and is used for automation and CI/CD integration.

Node.js

Node.js is required to install and run Newman. It provides the runtime environment for executing JavaScript-based tools.

ReqRes API

ReqRes is the testing API used in this project. It provides endpoints for user authentication and CRUD operations.

1.5 Scope of the Project

The scope of this project includes testing multiple API functionalities such as:

- User Login (Authentication testing)
- Fetching all users (GET request)
- Fetching a single user by ID
- Creating a new user (POST request)
- Updating user information (PUT and PATCH requests)
- Deleting a user (DELETE request)

Additionally, the project also includes:

- Validation of response status codes
- Verification of response body data

- Checking response time performance
 - Writing automated test scripts in Postman
 - Executing collections using Collection Runner and Newman
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1.6 Importance of API Testing

API testing is a very important part of software testing because:

- APIs are the core communication layer between frontend and backend
- Bugs in APIs can break the entire application
- It ensures data integrity and security
- It improves system reliability and performance
- It helps detect issues early in the development lifecycle

In modern software development, API testing is considered more important than UI testing in many cases because APIs handle business logic directly.

1.7 Real-Time Application of This Project

This project reflects real-world QA responsibilities such as:

- Testing backend services before deployment
- Validating data exchange between systems
- Ensuring APIs are reliable for mobile and web applications
- Running automated test suites in CI/CD pipelines
- Generating test execution reports for stakeholders

This makes the project highly useful for internships, interviews, and practical QA experience.

1.8 Conclusion of Introduction

The ReqRes API Testing Project provides a complete understanding of how API testing is performed using Postman and Newman. It helps build strong foundational knowledge of REST API testing, request-response validation, and automation execution.

By completing this project, a tester gains practical experience in API validation techniques that are widely used in the software testing industry.
